

11.1

$$A = \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \end{pmatrix}$$

hľadáme singulárny rozklad A^T

$$A^T = A^*$$

$$AA^* = \begin{pmatrix} 7 & -2 & -2 \\ -2 & 7 & -2 \\ -2 & -2 & 7 \end{pmatrix}$$

podľa Wolfram Alpha:

$$p_\lambda = -\lambda^3 - 21\lambda^2 + 135\lambda - 243 = -(\lambda - 9)^2(\lambda - 3) \quad \lambda_1 = 9 \quad \lambda_2 = 3$$

$$M_3 = \ker \begin{pmatrix} 4 & -2 & -2 \\ -2 & 4 & -2 \\ -2 & -2 & 4 \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} \quad d_1 = \sqrt{3}$$

$$M_9 = \ker \begin{pmatrix} -2 & -2 & -2 \\ -2 & -2 & -2 \\ -2 & -2 & -2 \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\} \quad d_{2,3} = 3$$

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \text{ tvoria bázu } \mathbb{R}^3$$

ortogonalizácia:

$$\vec{u}_1 = \frac{\sqrt{3}}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \vec{u}_2 = \frac{\sqrt{2}}{2} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

$$\text{Span} \{ \vec{u}_1, \vec{u}_2, \vec{u}_3 \} = \mathbb{R}^3$$

$$\vec{u}_3 = \frac{\sqrt{6}}{6} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$$

$$\vec{u}_1 \perp \vec{u}_2 \perp \vec{u}_3$$

$$\vec{u}_1 \cdot \vec{u}_3 = 0$$

$$\vec{u}_2 \cdot \vec{u}_3 = 0$$

$$U = \begin{pmatrix} \sqrt{3}/3 & -\sqrt{2}/2 & \sqrt{6}/6 \\ \sqrt{3}/3 & \sqrt{2}/2 & \sqrt{6}/6 \\ \sqrt{3}/3 & 0 & -\sqrt{6}/3 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 2\sqrt{3} & -3\sqrt{2} & \sqrt{6} \\ 2\sqrt{3} & 3\sqrt{2} & \sqrt{6} \\ 2\sqrt{3} & 0 & 2\sqrt{6} \end{pmatrix}$$

$$D = \begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix}$$

$$\vec{v}_1 = \frac{\sqrt{3}}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix} \cdot \frac{\sqrt{3}}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 3 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\vec{v}_2 = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix} \frac{\sqrt{2}}{2} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \frac{\sqrt{2}}{6} \begin{pmatrix} 0 \\ 0 \\ -3 \\ 3 \end{pmatrix} = \frac{\sqrt{2}}{2} \begin{pmatrix} 0 \\ 0 \\ -1 \\ 1 \end{pmatrix}$$

$$\vec{v}_1 \perp \vec{v}_2 \perp \vec{v}_3 \perp \vec{v}_4$$

$$\vec{v}_4 = \frac{\sqrt{2}}{3} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$v_1 \cdot v_4 = 0$$

$$v_2 \cdot v_4 = 0$$

$$v_3 \cdot v_4 = 0$$

$$\vec{v}_3 = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \\ -2 & 1 & 1 \end{pmatrix} \frac{\sqrt{6}}{6} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \frac{\sqrt{6}}{18} \begin{pmatrix} 0 \\ 6 \\ -3 \\ -3 \end{pmatrix} = \frac{\sqrt{6}}{6} \begin{pmatrix} 0 \\ 2 \\ -1 \\ -1 \end{pmatrix}$$

$$V = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{6}/3 & \sqrt{3}/3 \\ 0 & -\sqrt{2}/2 & -\sqrt{6}/6 & \sqrt{3}/3 \\ 0 & \sqrt{2}/2 & -\sqrt{6}/6 & \sqrt{3}/3 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 0 & 2\sqrt{6} & 2\sqrt{3} \\ 0 & -3\sqrt{2} & -\sqrt{6} & 2\sqrt{3} \\ 0 & 3\sqrt{2} & -\sqrt{6} & 2\sqrt{3} \end{pmatrix}$$

$$A = UDV^* = \frac{1}{6} \begin{pmatrix} 2\sqrt{3} & -3\sqrt{2} & \sqrt{6} \\ 2\sqrt{3} & 3\sqrt{2} & \sqrt{6} \\ 2\sqrt{3} & 0 & 2\sqrt{6} \end{pmatrix} \begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix} \frac{1}{6} \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 0 & -3\sqrt{2} & 3\sqrt{2} \\ 0 & 2\sqrt{6} & -\sqrt{6} & -\sqrt{6} \\ 0 & 2\sqrt{3} & 2\sqrt{3} & 2\sqrt{3} \end{pmatrix}$$

$$A^T = VD^*U^* = \frac{1}{6} \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 0 & 2\sqrt{6} & 2\sqrt{3} \\ 0 & -3\sqrt{2} & -\sqrt{6} & 2\sqrt{3} \\ 0 & 3\sqrt{2} & -\sqrt{6} & 2\sqrt{3} \end{pmatrix} \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix} \frac{1}{6} \begin{pmatrix} 2\sqrt{3} & 2\sqrt{3} & 2\sqrt{3} \\ -3\sqrt{2} & 3\sqrt{2} & 0 \\ \sqrt{6} & \sqrt{6} & 2\sqrt{6} \end{pmatrix}$$

$$AA^T = UDV^*VD^*U^*$$

$$V \text{ ortogonalna} \rightarrow VV^* = I_n \quad V^*V = I_n$$

$$AA^T = UDD^*U^*$$

$$DD^* = \begin{pmatrix} \sqrt{3} & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix}$$

$$AA^T = \frac{1}{6} \begin{pmatrix} 2\sqrt{3} & -3\sqrt{2} & \sqrt{6} \\ 2\sqrt{3} & 3\sqrt{2} & \sqrt{6} \\ 2\sqrt{3} & 0 & 2\sqrt{6} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} \frac{1}{6} \begin{pmatrix} 2\sqrt{3} & 2\sqrt{3} & 2\sqrt{3} \\ -3\sqrt{2} & 3\sqrt{2} & 0 \\ \sqrt{6} & \sqrt{6} & 2\sqrt{6} \end{pmatrix}$$

$$A^TA = VD^*U^*UDV^*$$

$$U \text{ ortog.} \rightarrow U^*U = I_n$$

$$A^TA = VD^*DU^*$$

$$D^* D = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \\ 0 & 0 & 0 \end{pmatrix}$$

$$A^T A = \frac{1}{6} \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 0 & 2\sqrt{6} & 2\sqrt{3} \\ 0 & -3\sqrt{2} & -\sqrt{6} & 2\sqrt{3} \\ 0 & 3\sqrt{2} & -\sqrt{6} & 2\sqrt{3} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \frac{1}{6} \begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 0 & -3\sqrt{2} & 3\sqrt{2} \\ 0 & 2\sqrt{6} & -\sqrt{6} & -\sqrt{6} \\ 0 & 2\sqrt{3} & 2\sqrt{3} & 2\sqrt{3} \end{pmatrix}$$